



BUS-1023 Applied Mathematics

CLO 2.1 Compute single trade  
discount, list price and net price.

Monday, September 21, 2026



| Terms                    | Definition                                                                                                  |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| 1. List Price            | The <b>list price</b> , or catalog price of an item, is the price for which you can generally sell an item. |
| 2. Net price             | The <b>net price</b> is the price you actually pay for the item.                                            |
| 3. Trade discount Amount | <b>Trade discount</b> is a discount off the list price                                                      |
| 4. Trade discount rate   | The <b>trade discount rate</b> is the amount of the discount expressed as a percentage                      |





# 1. List Price

The **Catalogue/  
Actual price**  
of a t-shirt is  
AED 240



## 2. Net Price

The **Final price** of a sofa set **after discount** is AED 599



**Sale Price**  
AED 599

After Sale Price AED 1,198  
Save AED 599



### 3. Trade Discount Amount

Grand optics are offering **\$100** off on all reading glasses





## 4. Discount Rate

There is **50% discount** on all white t-shirts.



### Example:

Home Center is on sale. Refer to their poster to find the following information about a study desk:

- a) List price = AED 1099
- b) Net price = AED 824
- c) Discount Rate = 25%
- d) Discount Amount = AED 275



**BESTSELLER** 25% OFF

Eden Study Desk

**AED 824** ~~AED 1099~~  
Save AED 275



Case 1: Find the **Net price** given the list price and trade discount amount.

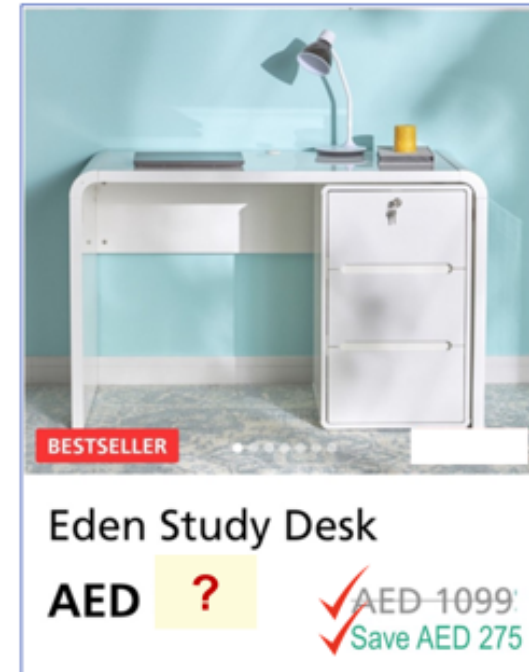
| NET PRICE = |   |                 |
|-------------|---|-----------------|
| List Price  | — | Discount Amount |

**Illustration:**

To find the net price of the desk:

Net Price = AED 1099 – AED 275

**Net Price = AED 824**





**Case 1: Find the net price given the list price and trade discount amount.**

**Example 1:** Home Décor is purchasing a motor for a clock. The **list price** of the motor is **\$499**. Home Décor receives **\$99 trade discount** from the wholesaler.

**Question :** What is the **Net price** of the motor?

**Answer :** The Net price = List Price – Trade Discount Amount

$$\text{The Net price} = \$499 - \$99 = \longrightarrow \$400$$



**Case 1: Find the net price given the list price and trade discount amount.**

**Example 2:** Howard bought stoves from a wholesaler. The **list price** of a stove is \$1270. Howard received **\$170 trade discount** from the wholesaler.

**Question :** What is the **Net price** of the stove?

**Answer :** The Net price = List Price – Trade discount Amount

$$\text{The Net price} = \$1270 - \$170 = \longrightarrow \$1100$$



Case 2: Find the **list price** given the net price and trade discount amount.

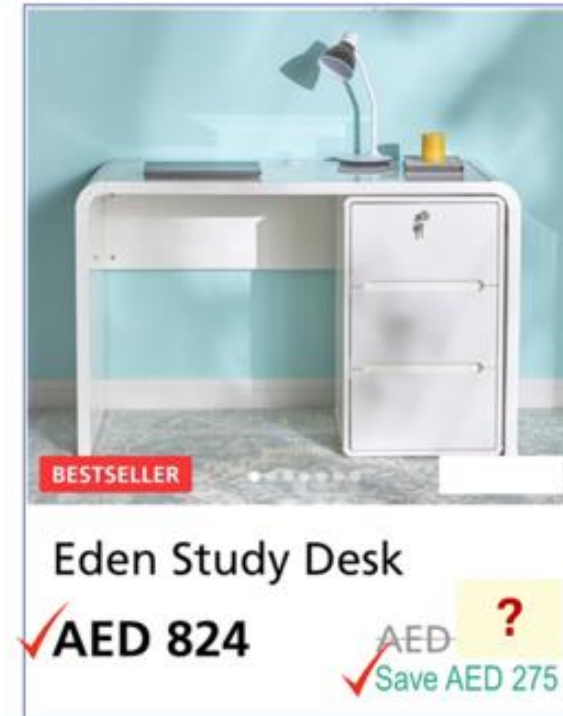
| LIST PRICE = |   |                 |
|--------------|---|-----------------|
| Net Price    | + | Discount Amount |

**Illustration:**

To find the list price of the desk:

List Price = AED 824 + AED 275

**List Price = AED 1099**



**Case 2: Find the list price given the net price and trade discount amount.**

**Example 1:** Home Décor is purchasing a motor for a clock. The **net price** of the motor is **\$400**. Home Décor receives **\$99 trade discount** from the wholesaler.

**Question :** What is the **list price** of the motor?

**Answer :** The list price = Net Price + Trade Discount Amount

$$\text{The list price} = \$400 + \$99 = \longrightarrow \$499$$



**Case 2: Find the list price given the net price and trade discount amount.**

1:

**Example 2:** Howard bought stoves from a wholesaler. The **net price** of a stove is **\$1100**. Howard received **\$170 trade discount** from the wholesaler.

**Question :** What is the **list price** of the stove?

**Answer :** The List price = List Price + Trade discount Amount

$$\text{The Net price} = \$1100 + \$170 = \text{—————} \$1270$$



Case 3: Find the **trade discount amount** given the list price and net price.

**TRADE DISCOUNT AMOUNT =**

**List Price**

**—**

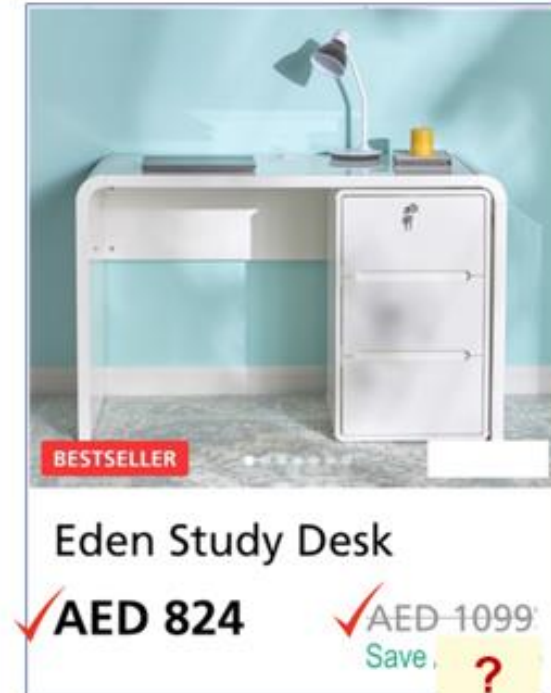
**Net Price**

**Illustration:**

To find the discount amount of the desk:

Trade discount amount = AED 1099 – AED 824

**Trade discount amount = AED 275**





**Case 3: Find the trade discount amount given the list price and net price.**

**Example 1:** Home Décor is purchasing a motor for a clock. The **net price** of the motor is **\$400**. The **list price** of the clock is **\$499**.

**Question :** Calculate the **trade discount amount**.

**Answer :** The trade discount amount = List Price – Net Price

$$\text{Trade Discount Amount} = \$499 - \$400 = \longrightarrow \$99$$



**Case 3: Find the trade discount amount given the list price and net price.**

**Example 2:** A washing machine has a **list price** of \$1000. The manufacturer offers its resellers a **net price** of \$850.

**Question :** Calculate the **trade discount amount**.

**Answer :** The trade discount amount = List Price – Net Price

$$\text{Trade Discount Amount} = \$1000 - \$850 = \longrightarrow \$150$$



Case 4: Find the **net price** given the list price and trade discount rate.

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### METHOD 1

**Step 1:** Calculate the **trade discount amount**.

TRADE DISCOUNT AMOUNT =

List Price **X** Discount Rate

**Step 2:** Calculate the **Net Price**.

NET PRICE =

List Price **—** Trade Discount Amount

OR

### METHOD 2

**Step 1:** Calculate the **complement of the discount rate**.

Complement of discount rate =

100% **—** Discount Rate

**Step 2:** Calculate the **Net Price**.

NET PRICE =

List Price **X** Complement of Discount Rate





#### Case 4: Find the **net price** given the list price and trade discount rate.

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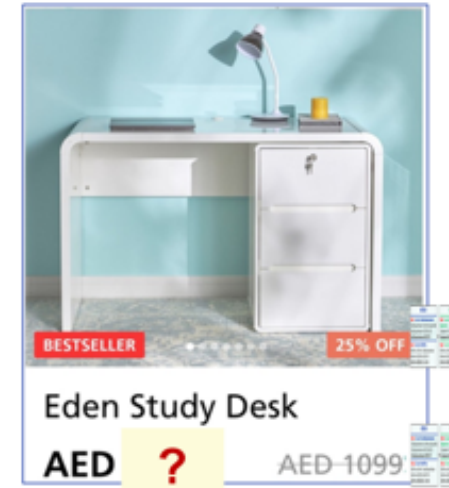
**Illustration:** To find the **net price** of the desk:

Given:

List Price = AED 1099

Trade Discount Rate = 25% or 0.25

Find: Net Price = ?



| METHOD 1                                                                                                                                                                                  |    | METHOD 2                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1: Calculate the trade discount amount.</b><br>Trade Discount Amount = List Price x Discount Rate<br>Trade Discount Amount = AED 1099 x 25%<br>Trade Discount Amount = AED 274.75 | OR | <b>Step 1: Calculate the complement of the discount rate.</b><br>Complement = 100% - 25%<br>Complement of Discount Rate = 75% or 0.75                    |
| <b>Step 2: Calculate the Net Price.</b><br>Net Price = List Price – Trade Discount Amount<br>Net Price = AED 1099 – AED 274.75<br>Net Price = AED 824.25 $\approx$ AED 824                |    | <b>Step 2: Calculate the Net Price.</b><br>Net Price = List Price x Complement<br>Net Price = AED 1099 x 75%<br>Net Price = AED 824.25 $\approx$ AED 824 |





**Case 4: Find the net price given the list price and trade discount rate.**

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**Example 1:** Home Décor is purchasing a motor for a clock. The **list price** of the motor is **\$499**. Home Décor receives a **10% trade discount** from the wholesaler.

**Question:** Calculate the **net price**. Then use the complement method to check your answer.

**METHOD 1**

**Step 1:** Calculate the **trade discount amount**.

Trade Discount Amount = List Price x Discount Rate

Trade Discount Amount = \$499 x 10%

**Trade Discount Amount = \$49.9**

**Step 2:** Calculate the **Net Price**.

Net Price = List Price – Trade Discount Amount

Net Price = \$499 – \$49.9

**Net Price = \$449.10**

**METHOD 2**

**Step 1:** Calculate the **complement of the discount rate**.

Complement = 100% – Trade Discount Rate

Complement = 100% – 10%

**Complement of Discount Rate = 90%**

**Step 2:** Calculate the **Net Price**.

Net Price = List Price x Complement

Net Price = \$499 x 90%

**Net Price = \$449.10**

**Same Answer**





**Case 4: Find the net price given the list price and trade discount rate.**

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**Example 2:** The catalogue price of a washing machine is \$1300. The manufacturer offers its resellers a 20% discount.

**Question:** Calculate the net price. Then use the complement method to check your answer.

**Answer:**

**Step 1 :**

**Trade Discount amount** = List Price x Trade discount rate = \$1300 x 20% = \$260

**Step 2 :**

**Net price** = List Price – Trade Discount amount = \$1300 – \$260 = \$1040







**Case 4: Find the net price given the list price and trade discount rate.**

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**Example 2:** The catalogue price of a washing machine is \$1300. The manufacturer offers its resellers a 20% discount.

**Question:** Calculate the net price using the complement method.

**Answer:**

**Step 1 :**

**Complement of the discount Rate** =  $100\% - \text{Trade discount rate} = 100\% - 20\% = 80\%$

**Step 2 :**

**Net price** = List Price x Complement of the Discount Rate =  $\$1300 \times 80\% = \$1040$

Same Answer





Case 5: Find the **list price** given the net price and trade discount rate.

$$\text{LIST PRICE} = \frac{\text{Net Price}}{\text{Complement of the Discount Rate}}$$

**Illustration:** To find the list price of the desk:

Complement of the Discount Rate = 100% – Discount Rate

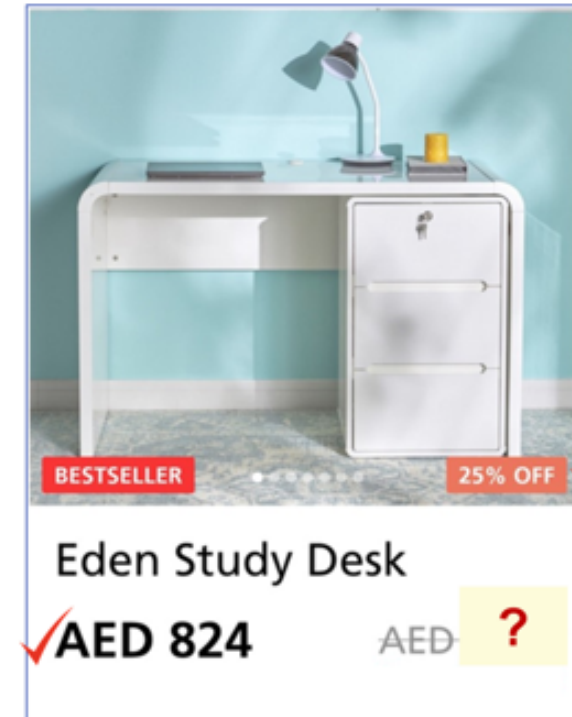
Complement of the Discount Rate = 100% – 25%

**Complement of the Discount Rate = 75%**

$$\text{List Price} = \frac{\text{Net Price}}{\text{Complement of the Discount Rate}}$$

$$\text{List Price} = \frac{\text{AED } 824}{75\%}$$

$$\text{List Price} = \text{AED } 1098.67 \approx \text{AED } 1099$$





**Case 5: Find the list price given the net price and trade discount rate.**

**Example 1:** Antique Wholesalers sells a clock motor at a **Net price** of **\$350**. The **trade-discount rate** is **5%**.

**Question:** Calculate the **list price**.

**Answer:**

**Step 1 :**

**Complement of the discount Rate** = 100% - Trade discount rate  
= 100% - 5%  
= **95%**

**Step 2 :**

List Price =  $\frac{\text{Net Price}}{\text{Complement of the Discount Rate}} = \frac{\$350}{95\%}$

**List Price = \$368.42**





**Case 5: Find the list price given the net price and trade discount rate.**

**Example 2:** A store bought a product from a manufacturer at a **net price** of **\$190**, and the **trade discount rate** is **75%**.

**Question:** Calculate the **list price**.

**Answer:**

**Step 1 :**

**Complement of the discount Rate** = 100% - Trade discount rate  
= 100% - 75%  
= **25%**

**Step 2 :**

$$\text{List Price} = \frac{\text{Net Price}}{\text{Complement of the Discount Rate}} = \frac{\$190}{25\%} = \$760$$





Case 6: Find the **trade discount rate** given the list price and net price

**Step 1:** Calculate the trade discount amount.

$$\text{TRADE DISCOUNT AMOUNT} = \text{List Price} - \text{Net Price}$$

**Step 2:** Calculate the Trade Discount Rate.

$$\text{Trade Discount Rate} = \frac{\text{Trade Discount Amount}}{\text{List Price}} \times 100$$





## Case 6: Find the **trade discount rate** given the list price and net price

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**Illustration:** To find the trade discount rate of the desk:

### **Step 1: Calculate the trade discount amount.**

Trade Discount Amount = List Price – Net Price

Trade Discount Amount = AED 1099 – AED 824

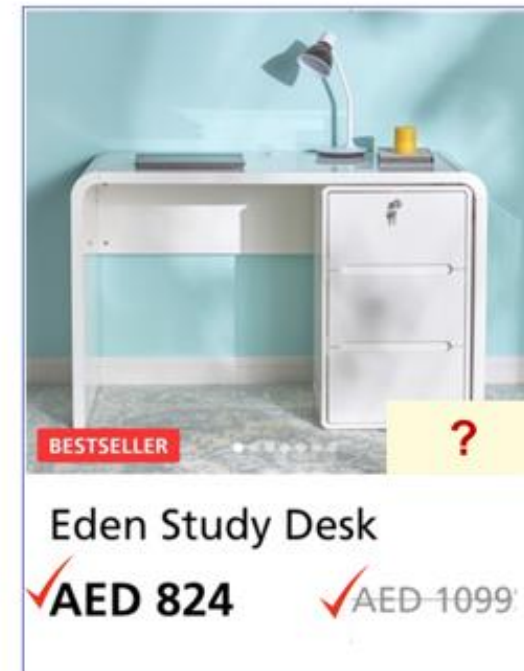
**Trade Discount Amount = AED 275**

### **Step 2: Calculate the Trade Discount Rate.**

$$\text{Trade Discount Rate} = \frac{\text{Trade Discount Amount}}{\text{List Price}} \times 100$$

$$\text{Trade Discount Rate} = \frac{\text{AED 275}}{\text{AED 1099}} \times 100$$

$$\text{Trade Discount Rate} = \boxed{25\%}$$



**Case 6: Find the trade discount rate given the list price and net price**

**Example 1:** Antique Wholesalers lists a clock motor in its **catalog** for **\$368.42**.

The **net price** to the retailer, Home Décor, is **\$350**.

**Question:** What is the trade discount rate?

**Answer :**

**Step 1 :**

**Amount of Discount** = List price – Net price = \$368.42 - \$350 = **\$18.42**

**Step 2 :**

$$\text{Trade Discount Rate} = \frac{\text{Trade Discount Amount}}{\text{List Price}} \times 100$$

$$\text{Trade Discount Rate} = \frac{\$18.42}{\$368.42} \times 100 = 4.9997\% \approx \mathbf{5\%}$$



**Case 6: Find the trade discount rate given the list price and net price**

**Example 1:** A computer has a **List Price** of \$550. The **Net price** to the retailers is \$400.

**Question:** What is the trade discount rate?

**Answer :**

**Step 1 :**

**Amount of Discount** = List price – Net price = \$550 - \$400 = **\$150**

**Step 2 :**

$$\text{Trade Discount Rate} = \frac{\text{Trade Discount Amount}}{\text{List Price}} \times 100$$

$$\text{Trade Discount Rate} = \frac{\$150}{\$550} \times 100 = \mathbf{27.3\%}$$





# Exercises:

Complete the following:

- [Exit Quiz 3.1](#) (Accessible once you have reviewed the PPT and videos)
- [Worksheet 3.1](#)



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